

Unseen Borders

Sprawl's Role in Fueling Las Vegas Inequality

Honors Thesis Defense by Riley Ramos

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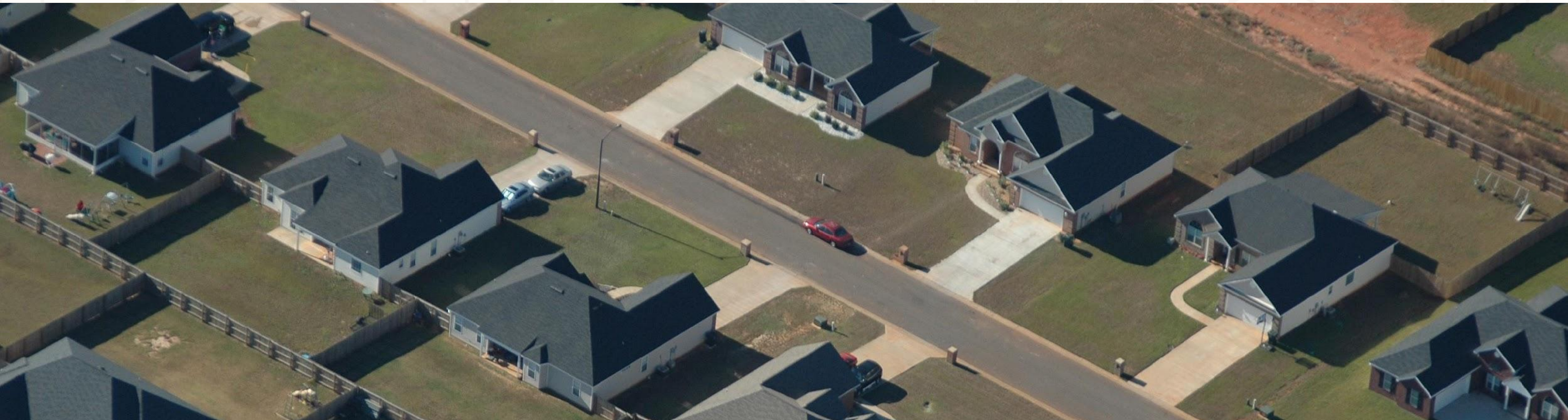
May 2025



UNLV

What is urban sprawl?

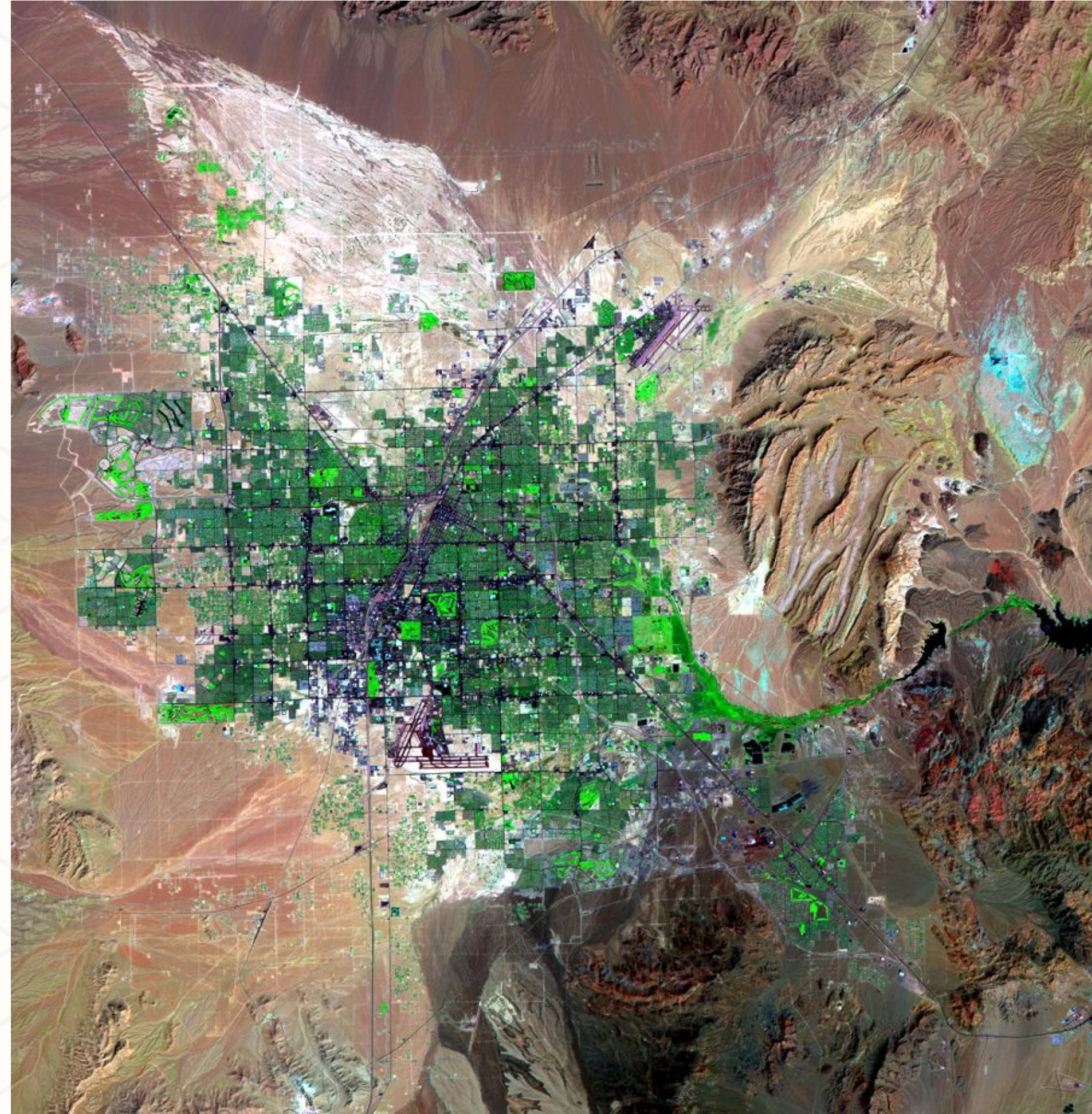
- Expansion of **peripheral** residential development.
- **Low** residential density.
- **Homogeneous** land use.
- **High** vehicle dependence.
- **Poor** street connectivity.



Rationale

Sprawl and inequality is understudied.

- Most studies examine sprawl's effect on land changes.
- Often conducted at the national scale.



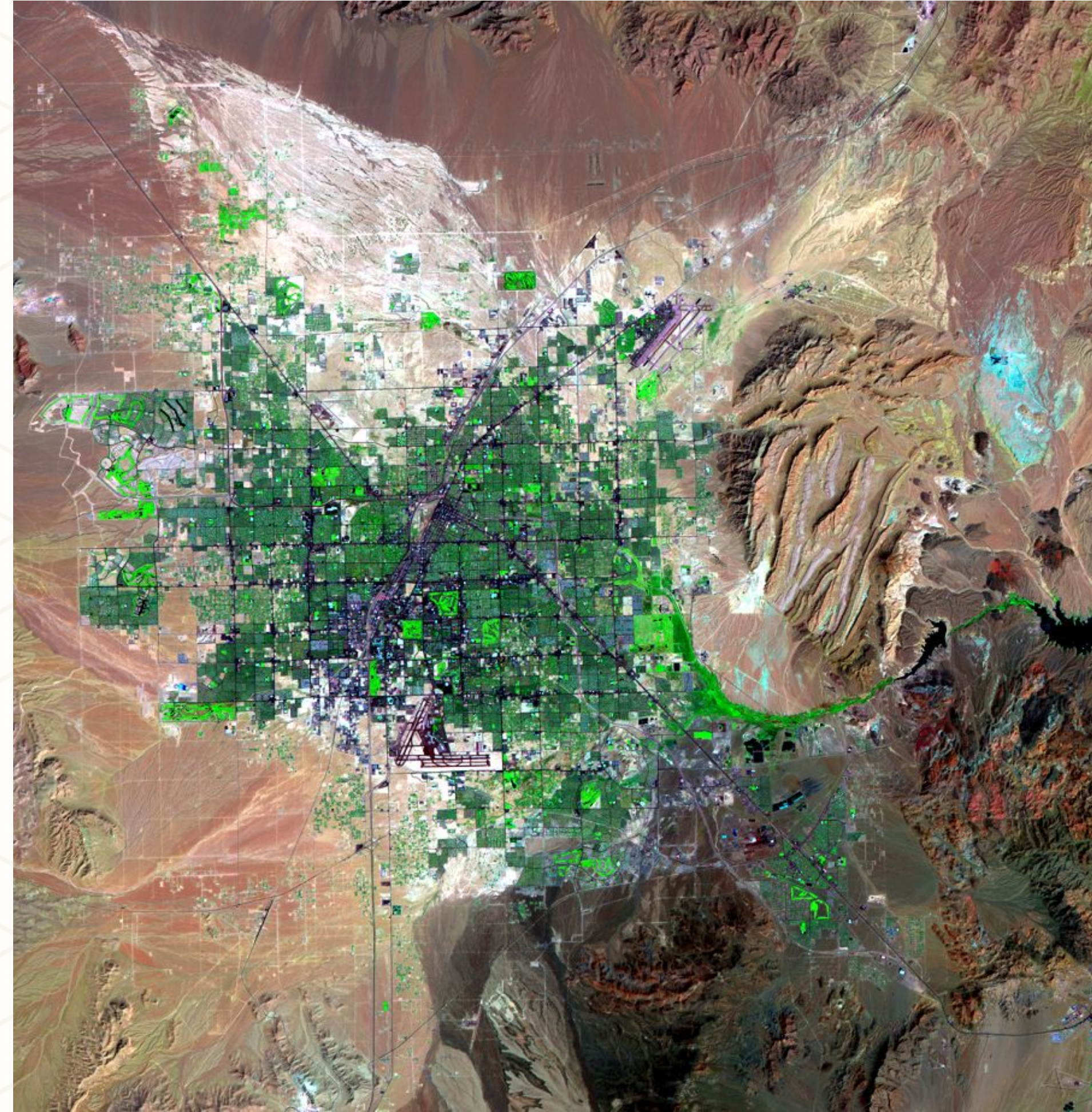
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Difficult to universally quantify sprawl

- Traditional metrics don't capture social inequality.
- Land and culture specific.



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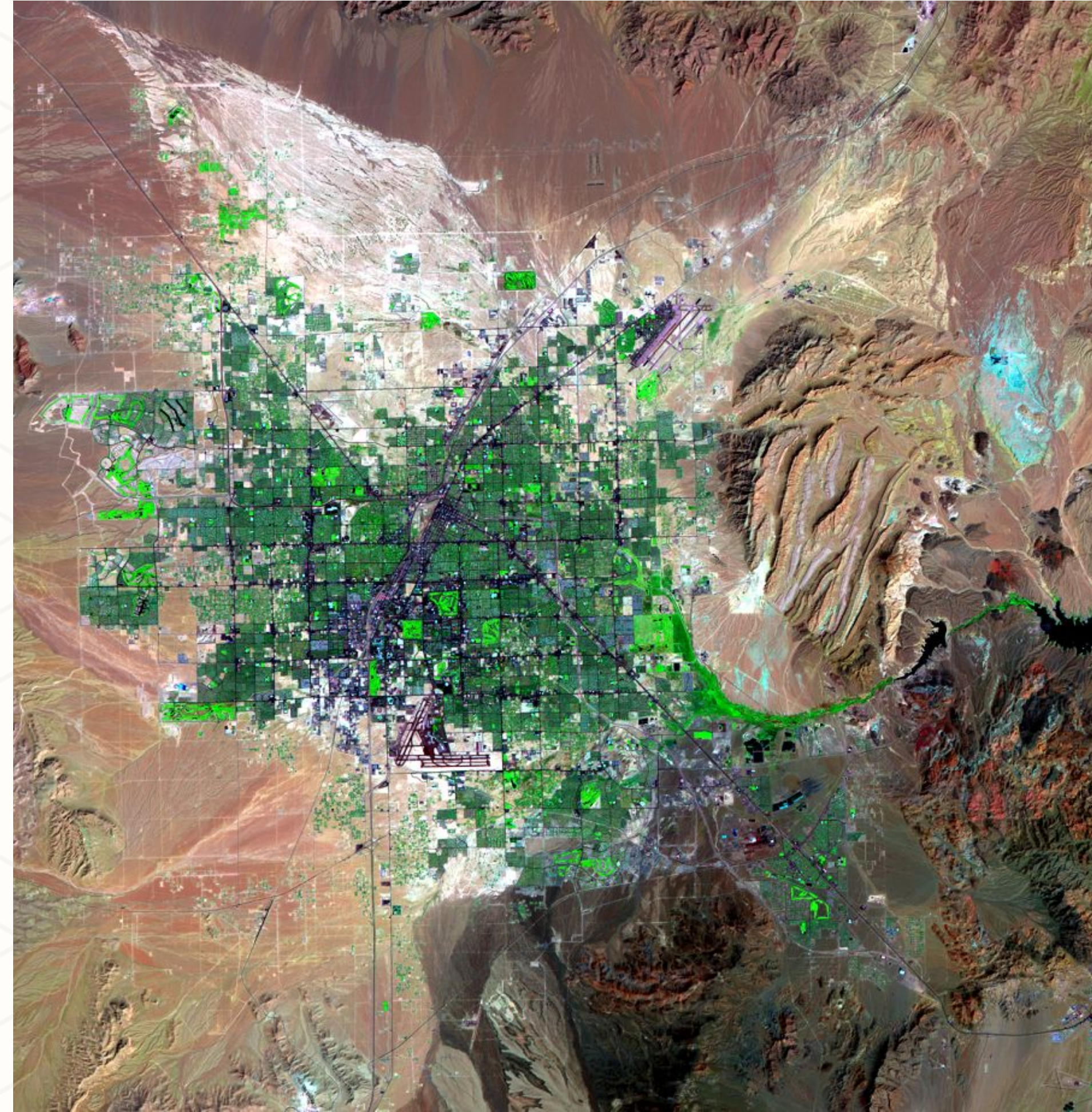
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Las Vegas sprawl is understudied.

- Rapid population growth since early 2000s.
- History of racial and income segregation.



Research Question & Hypothesis

Research Question

Are **people of color and low-income communities** disproportionately affected by sprawl outcomes?

Hypothesis

Sprawling areas have more **white and high-income** residents, while **less sprawling** areas tend to have more **racially marginalized and low-income** residents.

**** In my full thesis defense, I had **2 research questions** and **5 hypotheses** assessing sprawl's relationship with **commute, transportation, employment accessibility, and air pollution**. Stay tuned until the end to scan a QR code to view my full results!*

Data & Methods

Data

- **Urban sprawl index** (Ewing & Hamidi, 2010)
- **US Census** - demographics, commute, transportation, employment (2010)
- CDC daily **air pollution** (PM2.5) estimates

Methods

- **Spatial Maps**: displayed distribution of sprawl, people of color & low-income populations
- **Principal Component Analysis** (PCA): turned several sprawl-related factors into one combined “sprawl burden” score

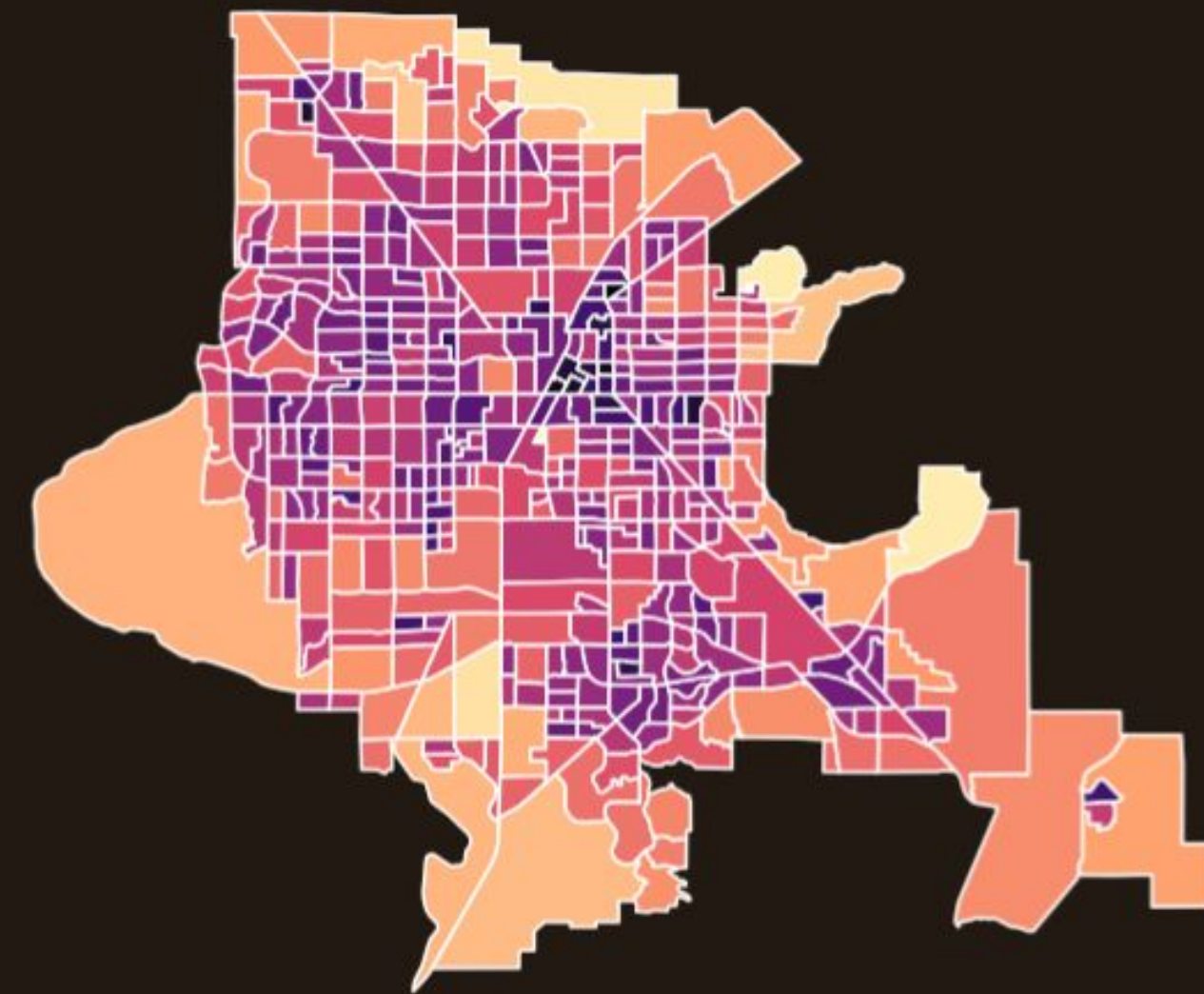
Other Methods (Not Included In This Presentation)

- Ordinary Least Squares (OLS) Linear Regression
- One-way Analysis of Variance (ANOVA)
- Summary statistics
- LASSO Coefficient Analysis

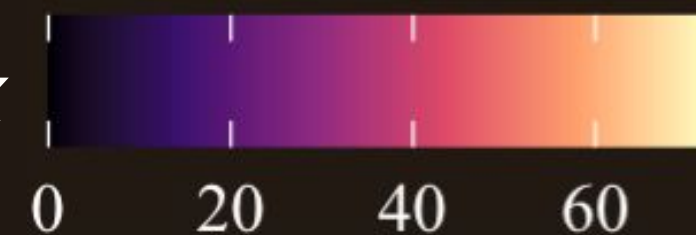
Results

- **Model:** Spatial Maps
- Sprawling census tracts located in the periphery
- Compact tracts in the center
- Exceptions in Henderson and other periphery areas
- **Generally supports** the hypothesis

Urban Sprawl in the Las Vegas Valley *Clark County, NV (2010)*



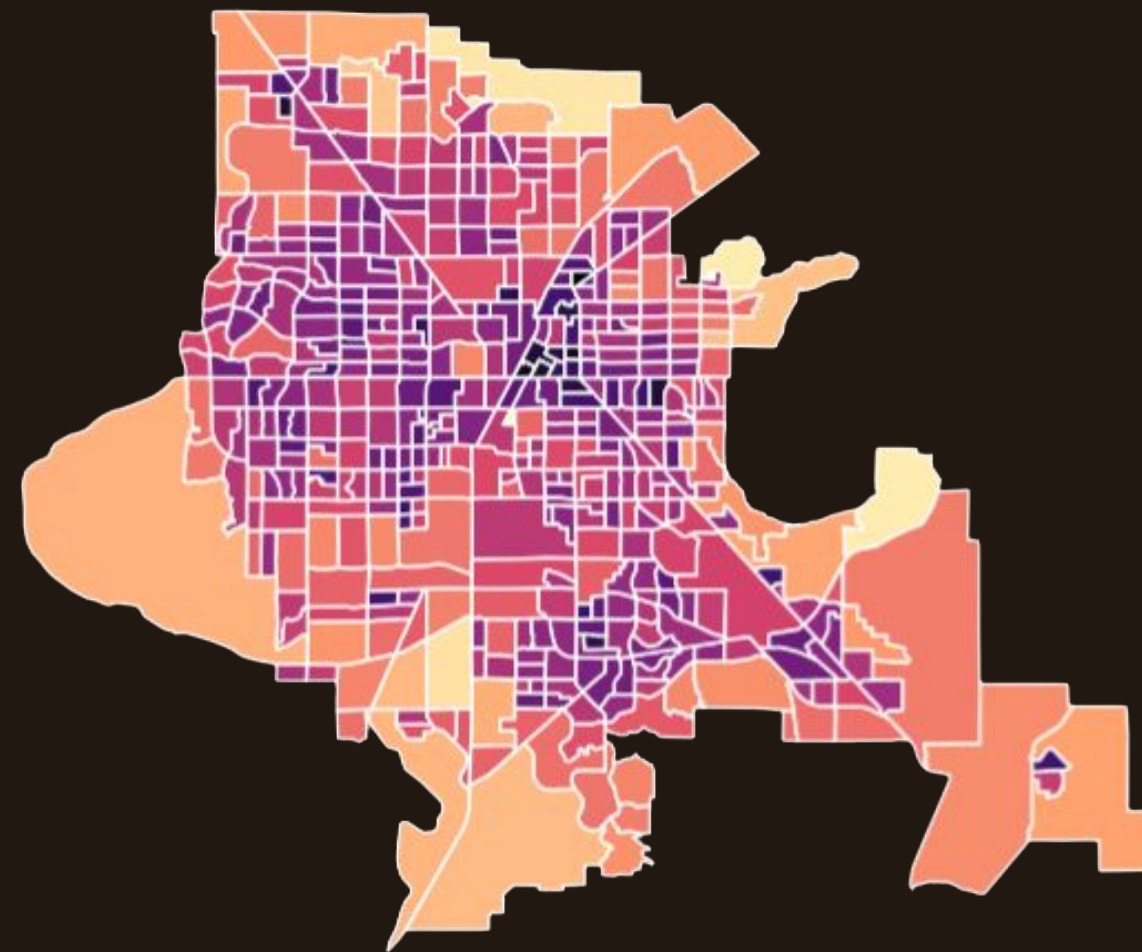
Sprawl Index



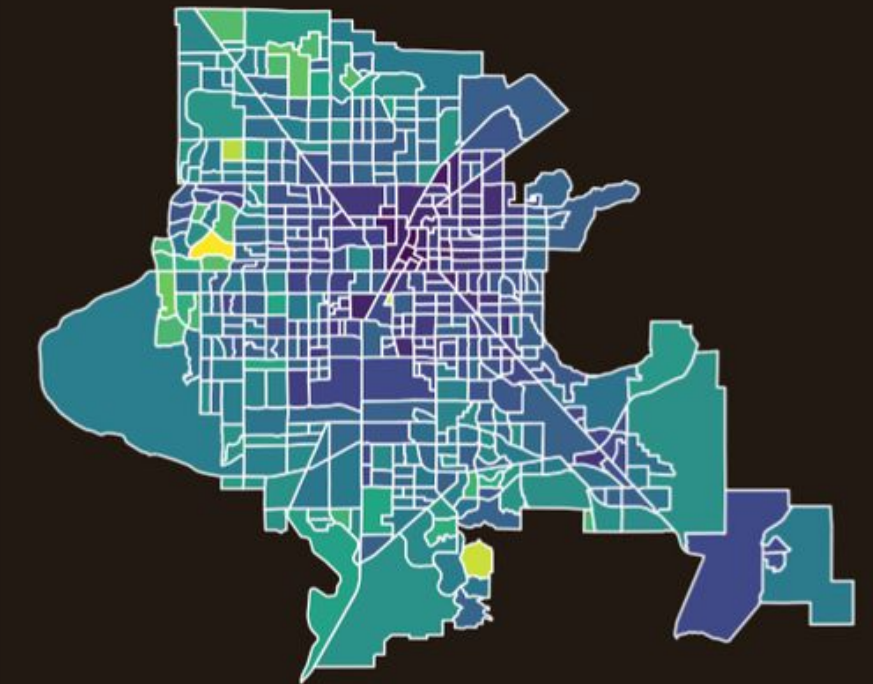
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Urban Sprawl



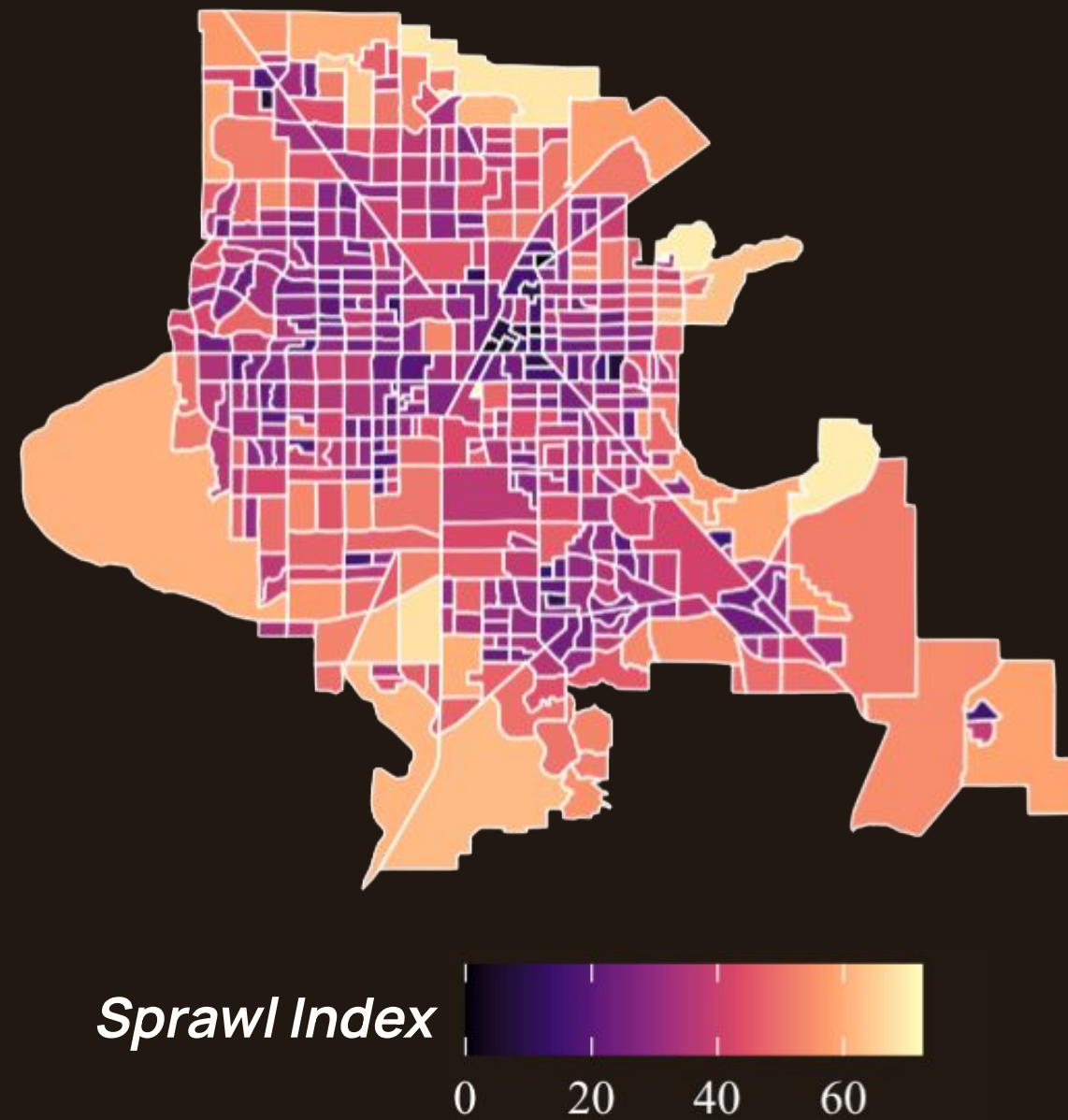
Median Income



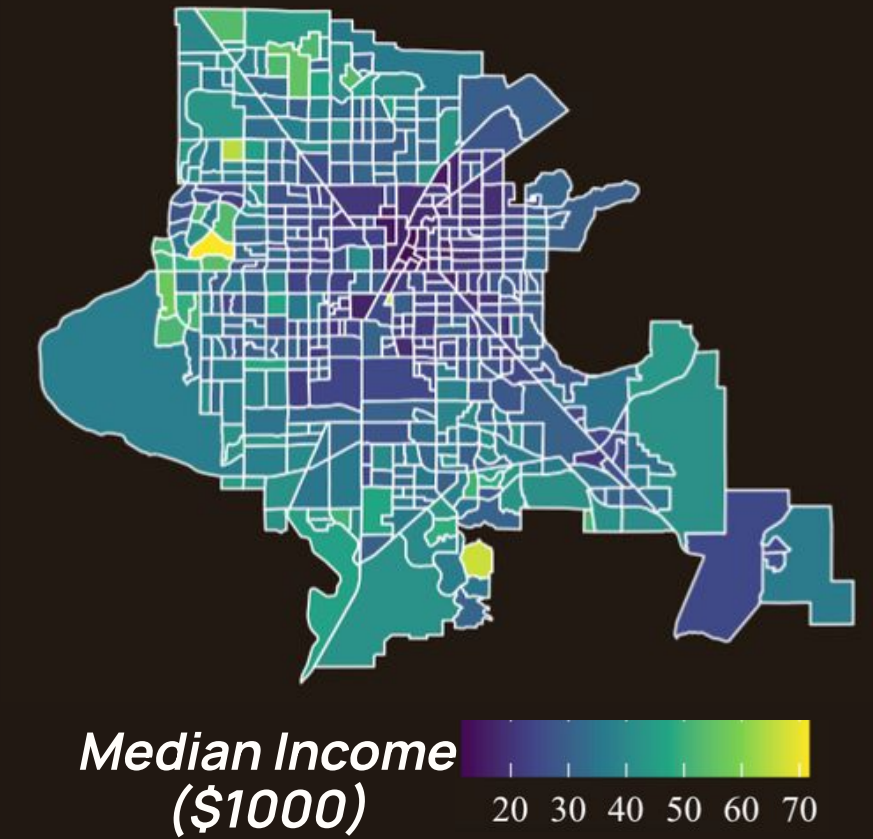
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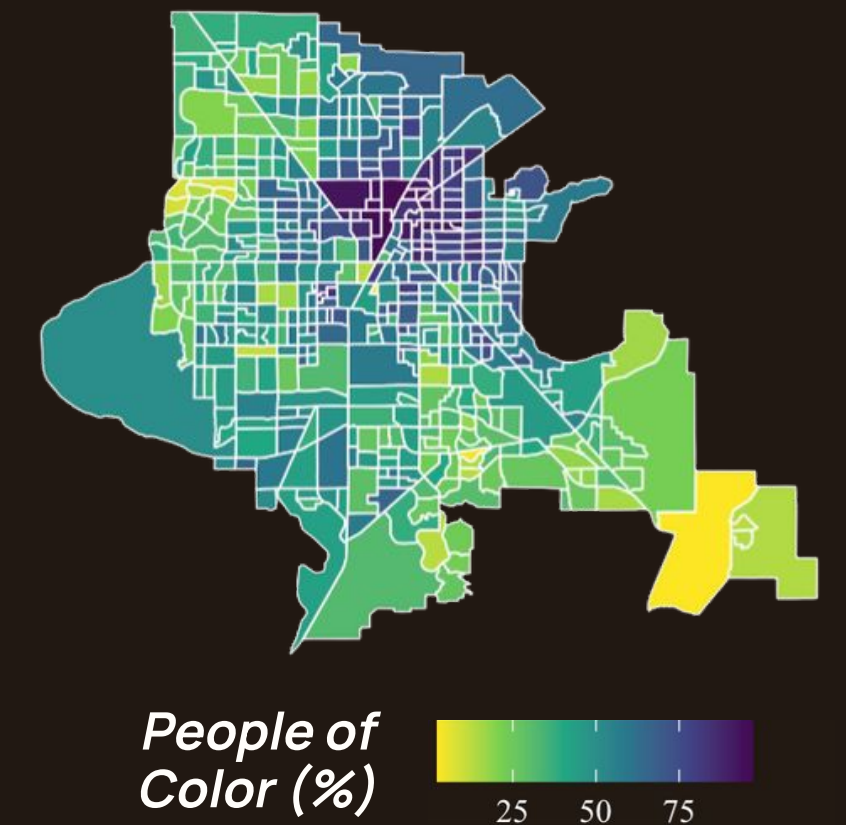
Urban Sprawl



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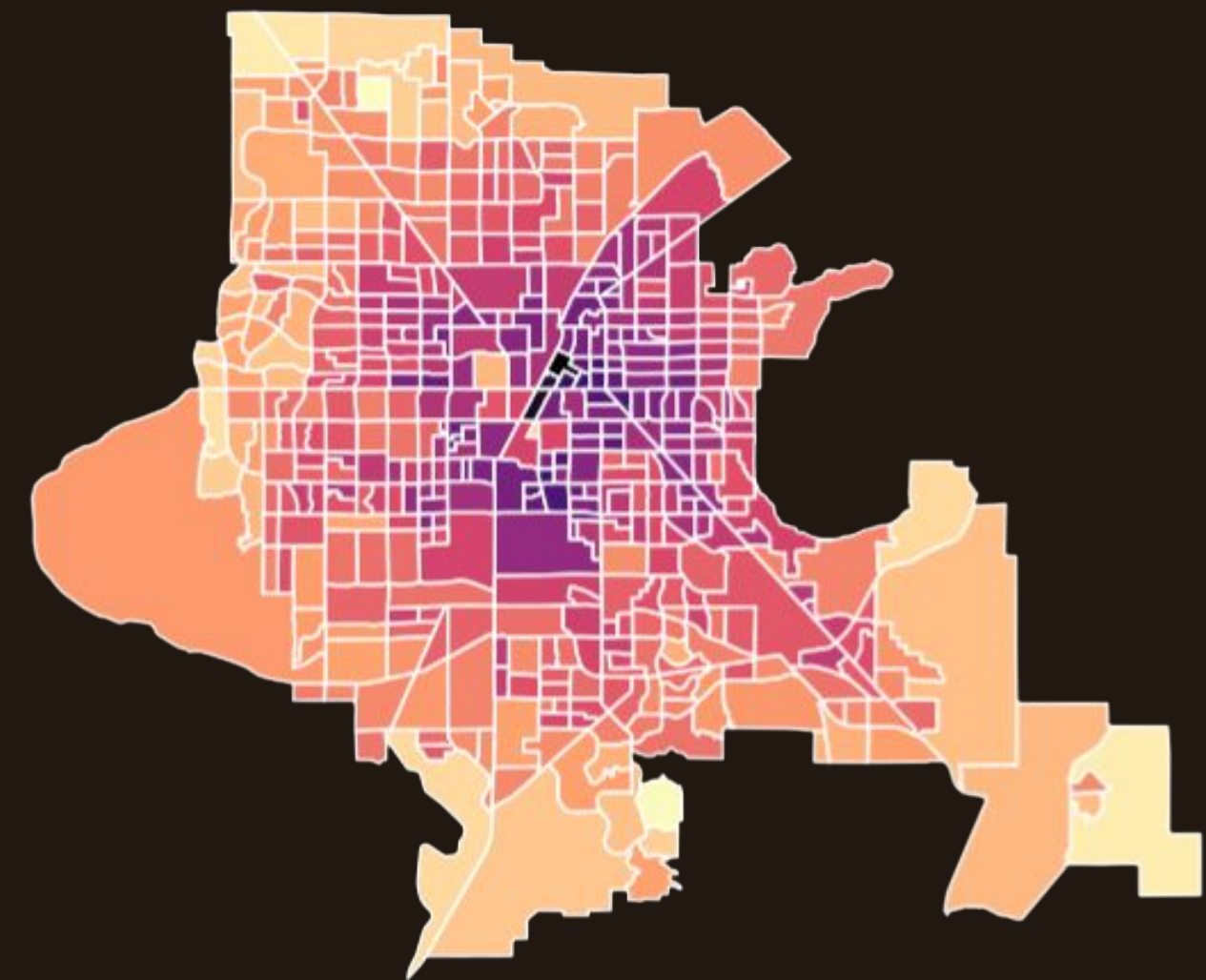
Percent People of Color



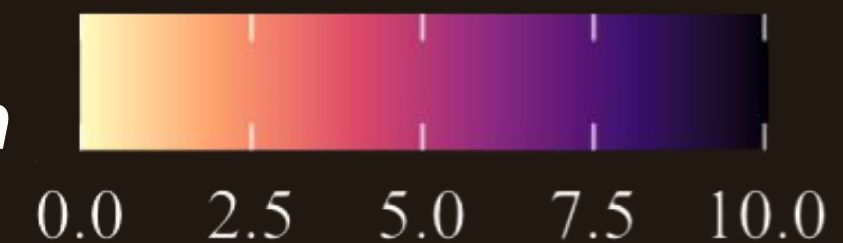
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- **Model:** Principal Component Analysis
- Census tracts that experience greater burden from sprawl are predominantly racially-marginalized and low income communities.
- **Affirms** research question

Sprawl Burden in the Las Vegas Valley *Clark County, NV (2010)*



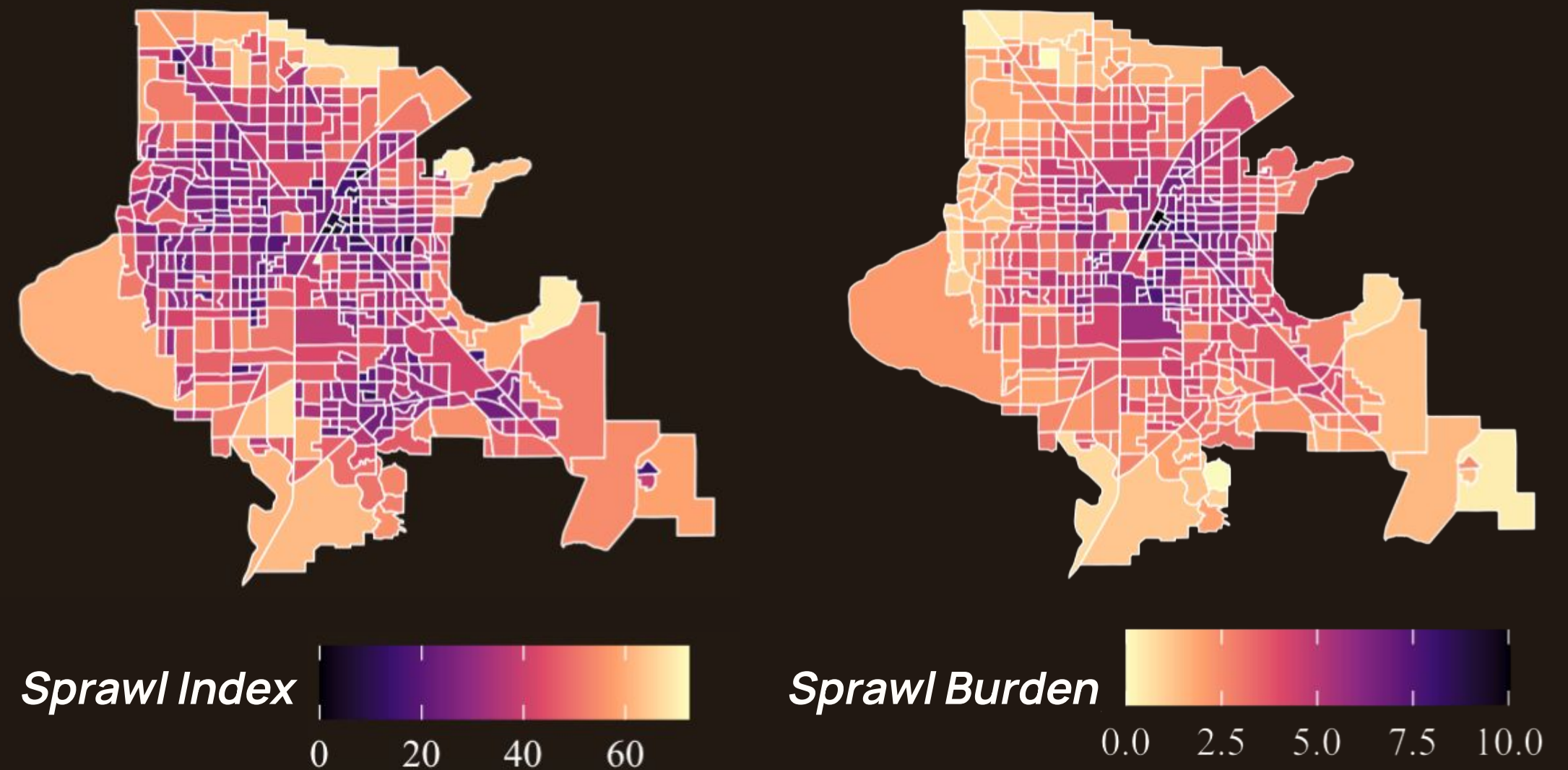
Sprawl Burden



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Sprawl Index vs. Burden in LV Valley *Clark County, NV (2010)*



Implications

- Sprawl's impact is more **nuanced** than what traditional metrics capture.
- Racially marginalized and low-income communities face **greater sprawl burden** in Las Vegas.
- Contributes to growing knowledge of the **harm inflicted by sprawl** in LV.



Thank You!

Questions?



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Scan to view my full
thesis defense presentation!



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